RAD: Rule-Augmented Relational Anomaly Detection

Noah Dahle, Anne Tumlin, Ngoc Tran, Xenofon Koutsoukos, Tyler Derr
Vanderbilt University, Nashville, TN, USA
CIKM '26 · Rome, Italy

Code & data

Key idea Mine symbolic rules from the data, inject them into target nodes ***before*** graph message passing, and train a heterogeneous graph autoencoder with a ranking loss — better anomaly retrieval in multi-table databases.

1.2
best average AUPRC rank of 10 methods across 3 benchmarks

0.659
AUPRC on LANL cyber events vs 0.525 for the best baseline

0.601
LANL precision@k: 60% of top alerts are real red-team logins (best baseline 0.540)

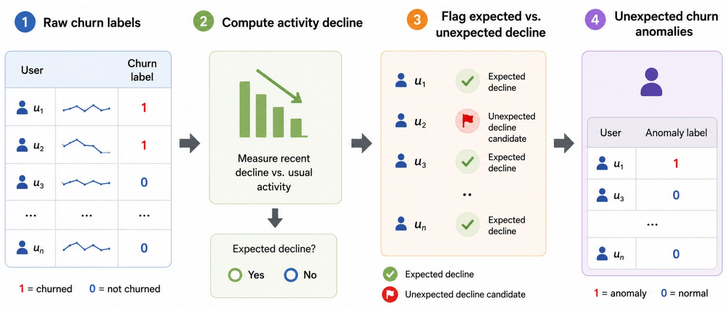
1 The Problem

A login can look normal as a single row yet be suspicious given who logged in, where, and what happened just before. Such anomalies live across tables.

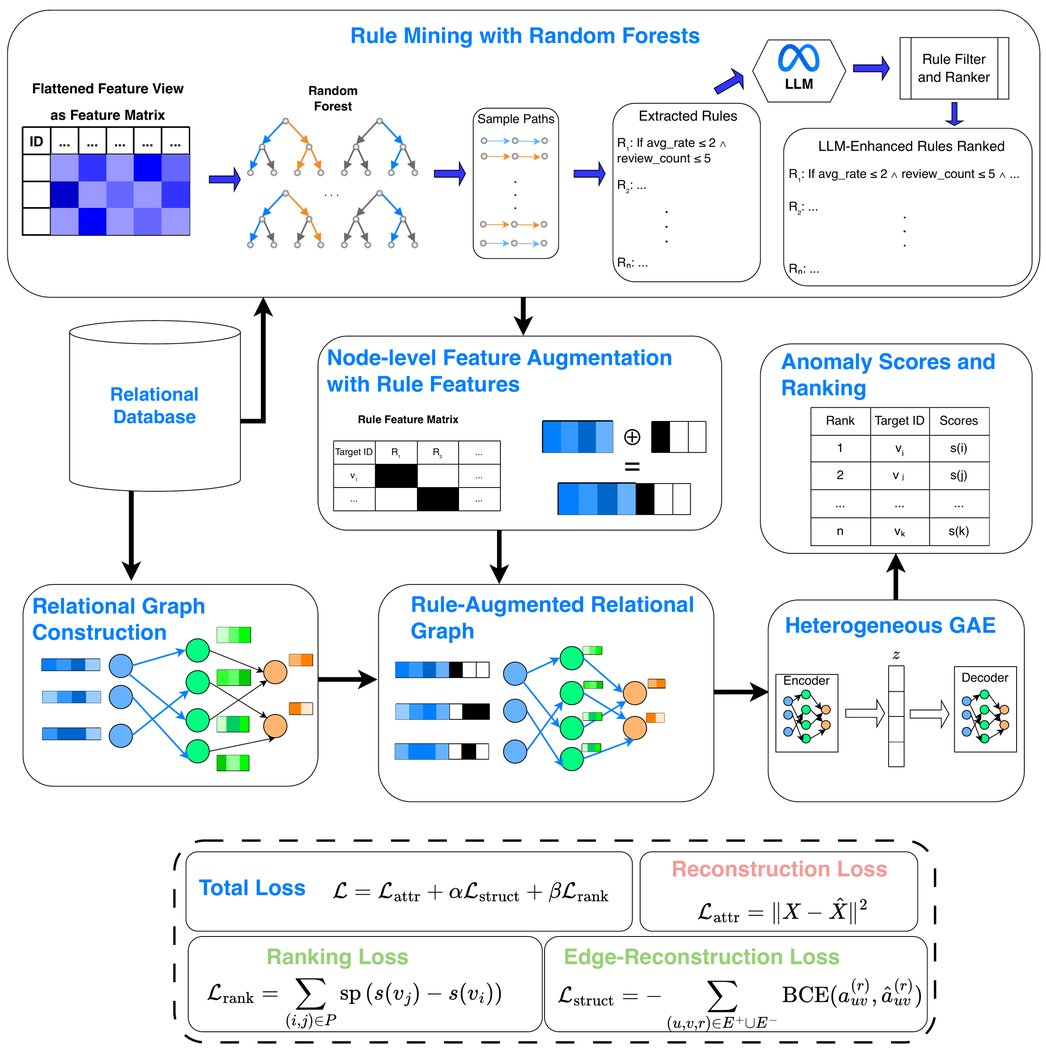
- Flattening into one feature matrix loses entity identity, schema roles and multi-hop context
- Relational deep learning keeps structure but has no anomaly objective
- Symbolic rules capture behavior, but usually sit outside representation learning

2 A New Benchmark

Task	Target	Records	Anomaly
LANL	Auth event	1.65B	Red-team login
Amazon	Customer	15.0M	Unexpected review churn
H&M	Customer	16.7M	Unexpected purchase churn



3 RAD: Rule-Augmented Relational Anomaly Detection



$$\textbf{Total Loss}\quad \mathcal{L} = \mathcal{L}_{\text{attr}} + \alpha\mathcal{L}_{\text{struct}} + \beta\mathcal{L}_{\text{rank}}$$

$$\textbf{Reconstruction Loss}\quad \mathcal{L}_{\text{attr}} = \|X - \hat{X}\|^2$$

$$\textbf{Ranking Loss}\quad \mathcal{L}_{\text{rank}} = \sum_{(i,j)\in P} \text{sp}\left(s(v_j) - s(v_i)\right)$$

$$\textbf{Edge-Reconstruction Loss}\quad \mathcal{L}_{\text{struct}} = -\sum_{(u,v,r)\in E^+\cup E^-} \text{BCE}(a_{uv}^{(r)}, \hat{a}_{uv}^{(r)})$$

1 Relational graph
Rows become typed nodes, primary–foreign keys become typed edges; only data available at time t is used.

2 Rule mining
Random-forest paths become conjunctive predicates, ranked by support, confidence and lift. An LLM prunes, tightens and merges them.

3 Rule injection
Each target node gets a binary rule vector r(v), concatenated to its features before message passing.

4 Rule-aware scoring
2-layer heterogeneous GraphSAGE autoencoder trained with reconstruction + pairwise ranking loss.

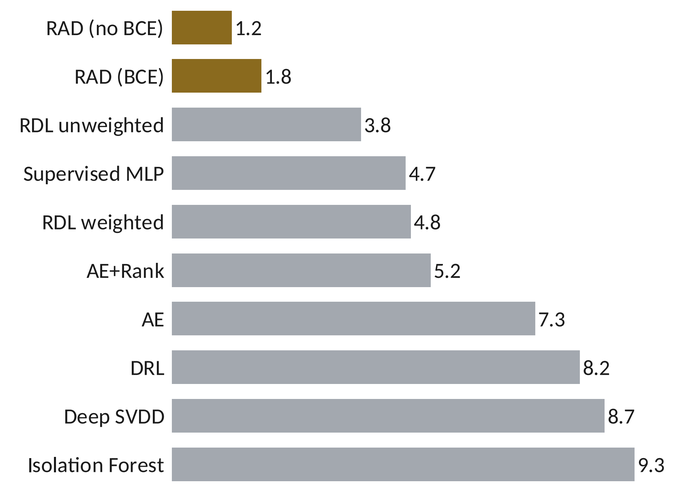
No label leakage. The LLM only proposes predicates, which are re-validated on train/validation data. It never assigns labels or scores.

4 Main Results (10 seeds)

Model	LANL			Amazon			H&M			Avg. rank ↓		
	AUROC	AUPRC	P@k	AUROC	AUPRC	P@k	AUROC	AUPRC	P@k	AUROC	AUPRC	P@k
RAD (rules, no BCE)	**0.996**	**0.659**	**0.601**	0.693	**0.043**	0.050	**0.828**	**0.143**	**0.129**	**2.0**	**1.2**	**2.0**
RAD (rules, BCE)	0.860	0.557	0.572	0.695	**0.043**	0.051	0.787	0.110	0.125	3.3	1.8	2.3
RDL weighted	0.553	0.007	0.021	**0.720**	0.037	**0.062**	0.700	0.035	0.064	4.7	4.8	3.7
RDL unweighted	0.685	0.015	0.026	0.713	0.037	**0.062**	0.693	0.039	0.064	4.3	3.8	3.3
AE+Rank	0.943	0.525	0.540	0.518	0.016	0.015	0.456	0.013	0.005	5.2	5.2	5.7
DRL	0.673	0.004	0.005	0.463	0.013	0.003	0.456	0.013	0.009	7.5	8.2	7.2
AE	0.888	0.009	0.000	0.474	0.014	0.014	0.408	0.012	0.007	6.7	7.3	7.8
Deep SVDD	0.736	0.005	0.001	0.458	0.013	0.003	0.412	0.012	0.002	7.7	8.7	8.8
Isolation Forest	0.644	0.003	0.000	0.450	0.013	0.001	0.402	0.012	0.004	9.7	9.3	9.5
Supervised MLP	0.965	0.050	0.093	0.659	0.027	0.041	0.642	0.027	0.039	4.0	4.7	4.7

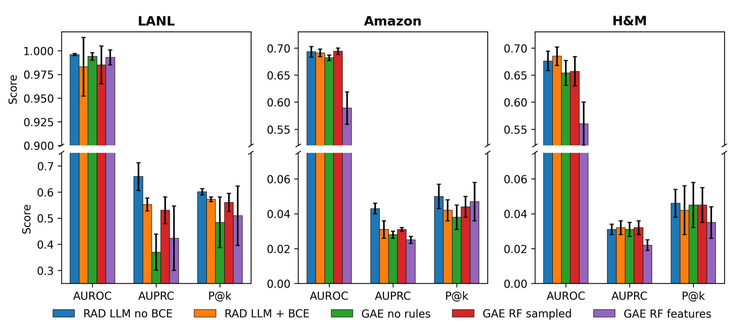
Means over 10 seeds (std. dev. in the paper). Bold = best per column. AUPRC and P@k matter most under heavy class imbalance.

5 Ranking at a Glance



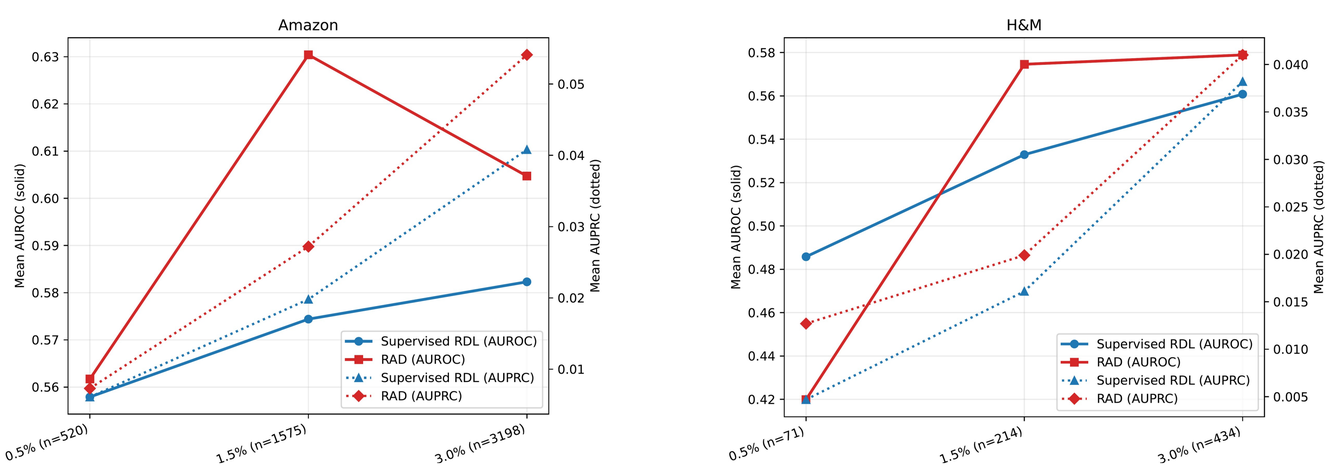
Average AUPRC rank across LANL, Amazon and H&M (lower is better). Both RAD variants rank ahead of every baseline, including the supervised MLP.

6 What Matters? Ablations



- Rules help: rule-augmented variants beat the no-rules GAE, most clearly on AUPRC
- Compact predicates beat raw random-forest decision features
- Edge reconstruction (BCE) is optional: the no-BCE variant ranks best overall

7 Robust to the Anomaly Definition



Varying the unexpected-churn prevalence (0.5%, 1.5%, 3.0%) leaves the RAD-vs-RDL comparison stable: the 1.5% default is not a cherry-picked threshold. Solid lines = AUROC, dotted = AUPRC.

Paper: doi.org/10.1145/3799682.3841086 **Code:** github.com/noahd15/RAD_RelationalAnomalyDetection Supported by NSA (H98230-23-C-0279) and NSF (IIS-2239881, ECCS-2325417, IIS-2524380, DGE-2622415).